CSE 2nd Year Even Semester Syllabus

Computer Science & Engineering • Rajshahi University of Engineering & Technology

CSE 2200

Technical Writing and Presentation Sessional

Contact Hours/Week: 3/2 Hours/Week | Credit Hour: 0.75

Introduction: Issues of Technical Writing and Effective Oral Presentation in Computer Science and Engineering; Criteria for Good Technical Writing and Presentation; An Overview of Research Methodologies: Quantitative and Qualitative Research.

Writing Issues: Writing Styles of Definitions, Propositions, Theorems, Proofs, etc.; Preparation of Reports, Scientific Articles, and Research Papers.

Thesis, Books and Others: Abstract, Preface, Contents, Bibliography and Index; Writing of Book Reviews and Referee Reports; Curriculum Vitae, Resume Writing, etc.

Writing and Presentation Tools: LATEX; Diagram Drawing Software; Presentation Tools.

Writing Ethics: Plagiarism Issues; Writing and Presentation Ethics; Ethical Technical Communication.

CSE 2201

Algorithm Analysis and Design

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Methods of Complexity Analysis: Substitution method, Iteration Method, Master Method, Amortized Analysis.

Linear Time Sorting: Radix Sort, Bucket Sort, Counting Sort, Proof of Correctness of Sorting algorithms, Lower bound on sorting.

Divide and Conquer approach: Minimum and Maximum Elements Finding, Median Finding, Kartasuba multiplication: Integer Multiplication, Strassens Matrix Multiplication, Quick Hull, Divide and Conquer Based Sorting Algorithms Review.

Greedy Algorithms: Greedy Algorithm Properties, Job Sequencing Problem, Huffman Codes, Fractional Knapsack Problem, Minimum Spanning Tree: Kruskal and Prims Algorithm, Single Source Shortest Path: Dijkstra’s Algorithm, Proof of correctness.

Dynamic Programming: Dynamic Algorithm Properties, Longest Common Subsequence (LCS), 0-1 Knapsack, Knapsack with duplicate items, Max Sum without Adjacent, Matrix Chain Multiplication (MCM), Single Source Shortest Path: Bellman-Ford Algorithm, All pair of shortest path: Floyd Warshall Algorithm, Bitmasking with DP: Traveling Salesperson Problem.

Backtracking: Backtracking, State Space Tree, N-Queen Problem, M-Coloring, Rat in a Maze, Solving Sudoku / Puzzle.

Branch and Bounds: Branch and Bounds in State Space Tree, LC Branch and Bound, FIFO Branch and Bound.

Flow Networks: Maximum Flow, Minimum Cut problem, Ford-Fulkerson Algorithm.

P and NP Problems: Introduction to Optimization and Decision Problems, Tractable and Intractable Problems, Deterministic and Nondeterministic Algorithms, P and NP Set of problems, NP-hardness and NP-completeness, Proof of NP-completeness for algorithms, Cook's Theorem.

Reducibility between Problems and Approximation Algorithms: Discussion of different NP-complete problems like Satisfiability, Clique, Vertex Cover, Independent Set, Hamiltonian Cycle, TSP and Knapsack. Reduction between problems, Approximation algorithm design for sample problems and corresponding performance bound analysis.

CSE 2202

Algorithm Analysis and Design Sessional

Contact Hours/Week: 3 Hours/Week | Credit Hour: 1.50

Sessional on Algorithm Analysis and Design: Methods of Complexity Analysis, Linear Time Sorting, Divide and Conquer Approach, Greedy Algorithms, Dynamic Programming, Backtracking, Branch and Bounds, Flow Networks, P and NP Problems, Reducibility between Problems and Approximation Algorithms.

CSE 2203

Numerical Methods

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Modeling, Computers and Error Analysis: Mathematical Modeling and Engineering Problem Solving, Programming and Software, Approximations and Round-Off Errors, Truncation Errors and the Taylor Series.

Roots of Equations: Bracketing Methods, Open Methods, Roots of Polynomials.

Linear Algebraic Equations: Gauss Elimination, LU Decomposition and Matrix Inversion, Gauss-Seidel.

Optimization: One-Dimensional Unconstrained Optimization.

Curve Fitting: Least-Square Regression.

Interpolation: Interpolation with one and two Independent Variables, Formation of Different Difference Table, Newton's Forward and Backward Difference, Langrange's Interpolation.

Numerical Differentiation and Integration: Newton-Cotes Integration Formulas, Integration of Equations, Numerical Differentiation.

Ordinary Differential Equations: Runge-Kutta Methods, Boundary-Value and Eigenvalue Problems, Numerical Solution of Partial Differential Equations.

CSE 2204

Numerical Methods Sessional

Contact Hours/Week: 3/2 Hours/Week | Credit Hour: 0.75

Sessional on Numerical Methods: Modeling, Computers and Error Analysis, Roots of Equations, Linear Algebraic Equations, Optimization, Curve Fitting, Interpolation, Numerical Differentiation and Integration, Ordinary Differential Equations.

CSE 2205

Microprocessors, Microcontrollers and Assembly Language

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Microprocessor: Introduction to Different Types of Microprocessors and its Applications, Organization of Intel x86 Architecture, Overview of Different Generations of Intel Processors, Component of Microcomputer System, Interrupt Structures, I/O Interfacing, DMA, Co-processors, Memory Module, Overview of Advanced Intel Instructions: AVX, MMX, SSE & SIMD.

Microprocessor Assembly: Introduction to x86 and x86-64 Assembly, Assembly Program Structure and Its Components, Basic Instructions, Input/Output Instructions, Flag Register, Flow Control Instructions, Logic, Shift and Rotate Instruction, Arithmetic Instructions, Arrays and Related Addressing Modes, Basic Stack Operations, Procedures Declaration, Calling a Procedure, String Instructions, Macros, Floating-Point Instructions.

Microcontroller: Microcontroller History and Features, Microcontroller Architecture, General Purpose Registers in Microcontroller, Microcontroller Data Memory, Microcontroller Status Register, Microcontroller Data Format and Directives.

Microcontroller Assembly: Microcontroller Assembly Programming, Program Counter and Program ROM Space in Microcontroller, Branch Instructions and Looping, Call Instructions and Stack, Time Delay, Arithmetic Instructions, Logic and Compare Instructions, Rotate and Shift Instructions.

CSE 2206

Microprocessors, Microcontrollers and Assembly Language Sessional

Contact Hours/Week: 3 Hours/Week | Credit Hour: 1.50

Sessional on Microprocessors, Microcontrollers and Assembly Language: Environment Setup, Microprocessor Assembly: Registers and Program Structure, Flow Control, Looping and Branching, Logic and Arithmetic Operation, Stack Operation, Arrays and Data Structure, Addressing Modes, String Manipulation, Macros, Microcontroller Assembly: Branch and Looping Instructions, Call Instructions and Stack, Time Delay, Arithmetic Instructions, Logic and Compare Instructions, Rotate and Shift Instructions.

Math 2213

Complex Variable, Partial Differential Equation and Harmonic Analysis

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Complex Variable: Complex Number Systems, General Functions of a Complex Variable, Limits and Continuity of a Function of Complex Variable and Related Theorems, Complex Differentiation and the Cauchy-Riemenn Equations, Infinite Series, Convergence, Line Integral, Cauchy Integral Theorem, Cauchy Integral Formula, Liouville's Theorem, Taylor's and Laurent's Theorems, Singular Points, Residue, Cauchy's Residue Theorem, Contour Integration.

Partial Differential Equation: Partial Differential Equations, Solution of First Order Partial Differential Equation by Lagrange and Charpit Methods, Solution of Laplace Equation and Wave Equation.

Harmonic Analysis and Laplace Transform: Fourier Series and Fourier Transformations and its Applications to Solve Boundary Value Problems. Laplace Transforms, Inverse Laplace Transforms, Solution of Differential Equation by Laplace Transforms.

Hum 2213

Industrial Management and Accountancy

Contact Hours/Week: 3 Hours/Week | Credit Hour: 3.00

Industrial Management: Principles of Management, Management Functions, Management Skills, Authority and Responsibility, Span of Control, Management by Objective, Consultative Management, Participative Management, Decision Making, Manpower Motivation, Human Resources Management: Manpower Planning, Recruitment and Selection, Employee Training and Development, Performance Appraisal, Wages and Salary administration, Production Management: Plant Layout: Definition, Basic Layout Types, Problem Solving, Linear Programming, EOQ, Lead Time, Safety Stock, Re-order Point.

Accounting: Basic Accounting Principles, Objectives of Accounting, Transaction, Double Entry Systems, Accounts and its Classification, Journals, Cash Book, Ledger, Trial Balance, Financial Statement, Cost Accounts and its Objectives, Cost Classification, Preparation of cost sheet, Cost Volume Profit analysis, Standard Costing, Process Costing.
